



Relating Health, Activity, and Obesity in the United States

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Data of Interest

Dataset Name

Nutrition, Physical Activity, and Obesity – Behavioral Risk Factor Surveillance System (BRFSS)

Organization

Collected and published by the Centers for Disease Control and Prevention (CDC).

Dataset URL

<https://catalog.data.gov/dataset/nutrition-physical-activity-and-obesity-behavioral-risk-factor-surveillance-system>

File Type

CSV format

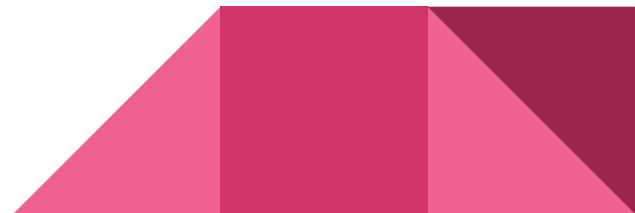


Important Dataset Variables

The dataset includes percentages related to:

- Adult obesity rates
- Physical inactivity
- Fruit/vegetable consumption
- Exercise participation
- Socioeconomic health indicators

Variable	Description
Location	U.S. State
Year	Year of data collected
Question	Health related survey question
Data_value	Measured percentage
Stratification	Demographic
Class	Health topic category
Topic	Specific health topic



How and why was the Data Collected?

Why This Dataset Was Collected

The CDC collected this data to monitor health trends related to:

- Obesity
- Physical activity
- Nutrition
- Chronic disease risk factors
- Public health outcomes across the United States

Why This Dataset Is Useful

This dataset allows analysis of:

- Obesity prevalence by state
- Exercise participation
- Access to healthy food
- Trends over time
- Differences among demographic groups



Potential Question 1

Question

How have obesity rates changed over time across the United States?

Why This Question Is Interesting

This question investigates historical obesity rates as well as developing a trendline to assist in predicting rates in future years

Variables of Interest

- Year
 - Obesity percentage
- 

Potential Question 2

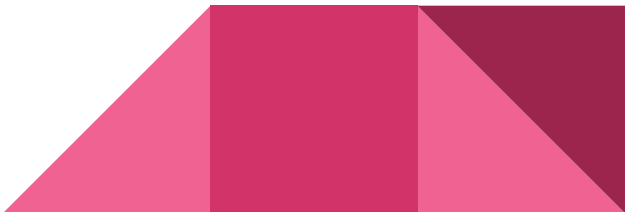
Question

How do physical activity levels and dietary habits compare in their relationship to obesity rates across different years?

Why This Question Is Interesting

This question compares two major lifestyle factors that may contribute to obesity and investigates which appears to have the stronger relationship.

Variables of Interest

- Physical inactivity percentage
 - Fruit and vegetable consumption percentage
 - Adult obesity percentage
 - State/location
- 

Potential Question 3

Question

Is there a relationship between obesity rates across different demographic groups

Why This Question Is Interesting

This question explores how health behaviors and income may influence obesity prevalence across different populations.

Variables of Interest

- Obesity percentage
- Demographic groups



Potential Question 4

Question

Which regions of the United States consistently show the highest obesity rates?

Why This Question Is Interesting

Regional differences may reveal environmental or socioeconomic influences.

Variables of Interest

- State/location
 - Obesity percentage
 - Geographic region
- 

Data and Question Alignment

Question	Dataset Features Used	Can this data answer the question?
How obesity changes over time	YearStart, Data_Value, Question	Yes
Relationship between physical activity, diet, and obesity	Question, Data_Value, LocationDesc	Yes
Relationship between demographic groups and obesity	Question, Data_Value, Stratification1, LocationDesc	Yes
Regional obesity differences	LocationDesc, Data_Value	Yes

Selected Question 1

Selected Question

How have obesity rates changed over time?

Proposed Analysis

- Calculate the national average year obesity rates
- Create a line of best fit and calculate the correlation coefficient

Expected Result

The data will show a strong correlation between an increase in years and an increase in obesity

Metrics

- Mean obesity rate per year
- Rate of annual increase

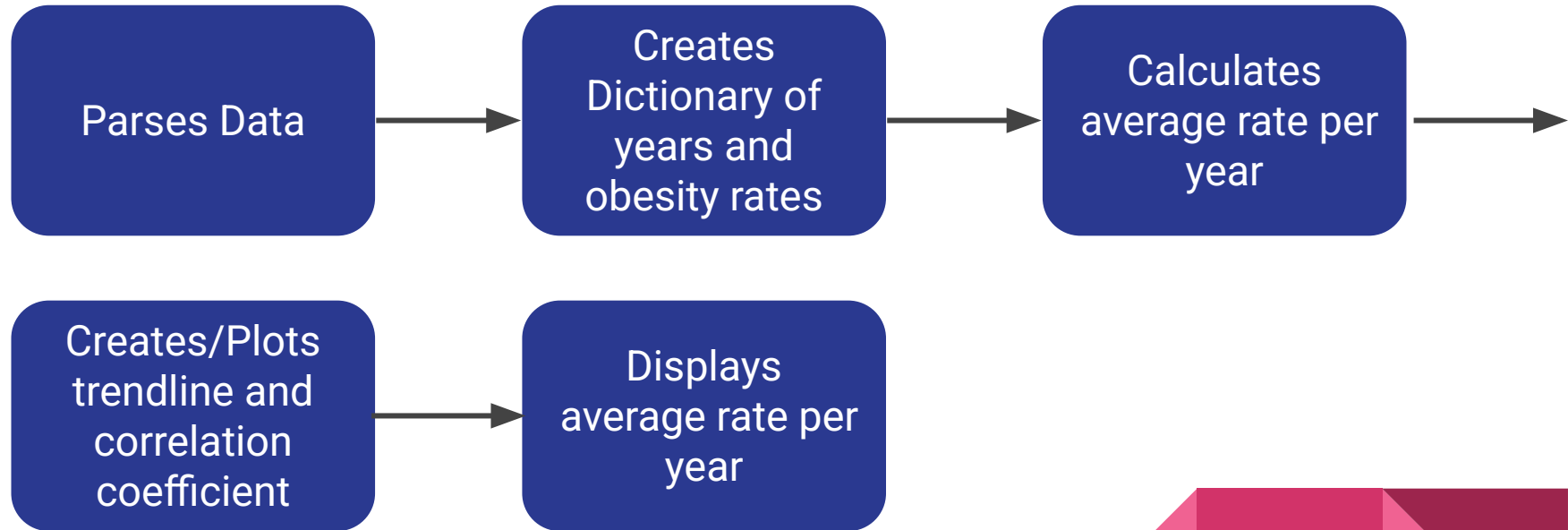
Python Packages

- pandas
- matplotlib
- numpy

Plots

- Time-series line graph

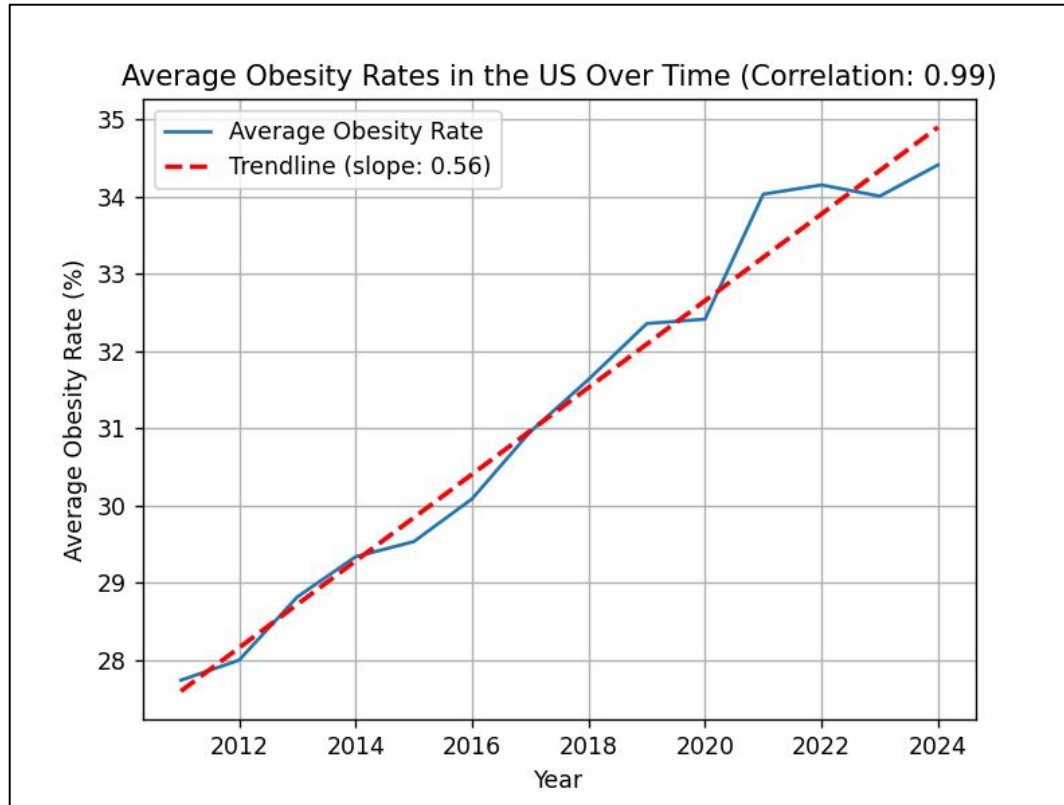
General Program Flow for Question 1



Question 1 – Command Line Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  +
(venv) PS C:\Users\Jacob\OneDrive - James Madison University\JMU\Spring '26\Computational Methods\GitHub\ENGR315-sp2026-studen
Year: 2011, Average Obesity Rate: 27.74%
Year: 2012, Average Obesity Rate: 28.00%
Year: 2013, Average Obesity Rate: 28.82%
Year: 2014, Average Obesity Rate: 29.34%
Year: 2015, Average Obesity Rate: 29.53%
Year: 2016, Average Obesity Rate: 30.09%
Year: 2017, Average Obesity Rate: 30.96%
Year: 2018, Average Obesity Rate: 31.63%
Year: 2019, Average Obesity Rate: 32.36%
Year: 2020, Average Obesity Rate: 32.41%
Year: 2021, Average Obesity Rate: 34.03%
Year: 2022, Average Obesity Rate: 34.15%
Year: 2023, Average Obesity Rate: 34.01%
Year: 2024, Average Obesity Rate: 34.41%
```

Plot for Question 1



Conclusions for Question 1

- With a correlation coefficient of 0.99 there is a clear trend between time and average obesity rates.
- A trendline slope of 0.56 leads us to predict a national average obesity rate of 40% in 2036.
- This analysis doesn't provide an explanation for the trend however, and that required further analysis.



Selected Question 2

Selected Question

How do physical activity levels and dietary habits compare in their relationship to obesity rates across different years?

Proposed Analysis

- Scatter plots comparing obesity vs physical inactivity
- Scatter plots comparing obesity vs fruit/vegetable consumption
- Correlation coefficient calculations
- Linear regression trendlines

Expected Result

Physical inactivity is expected to show a stronger positive relationship with obesity, while healthier diets are expected to show a negative relationship.

Metrics

- Average inactivity percentage
- Average obesity percentage
- Correlation coefficient

Python Packages

- pandas
- matplotlib
- numpy
- scipy

Plots

- Scatter plot
- Regression line

General Program Flow for Question 2



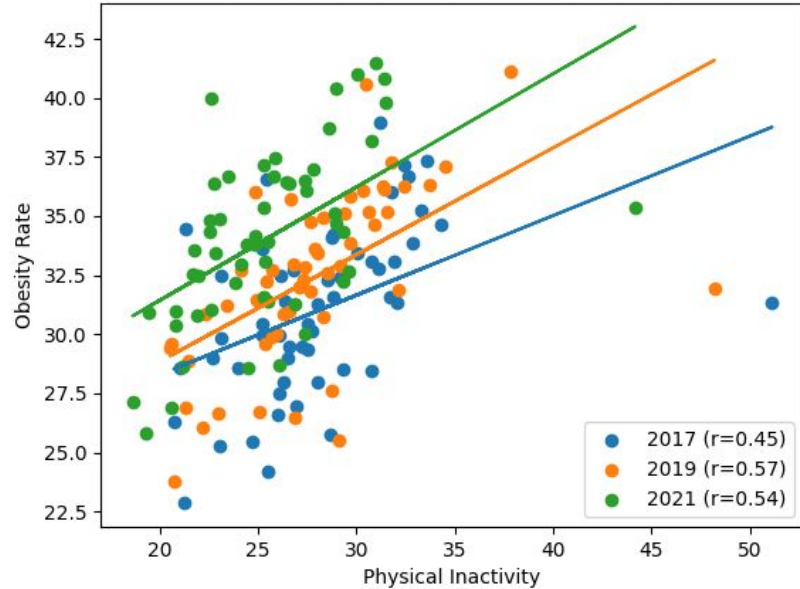
Question 2 – Command Line Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

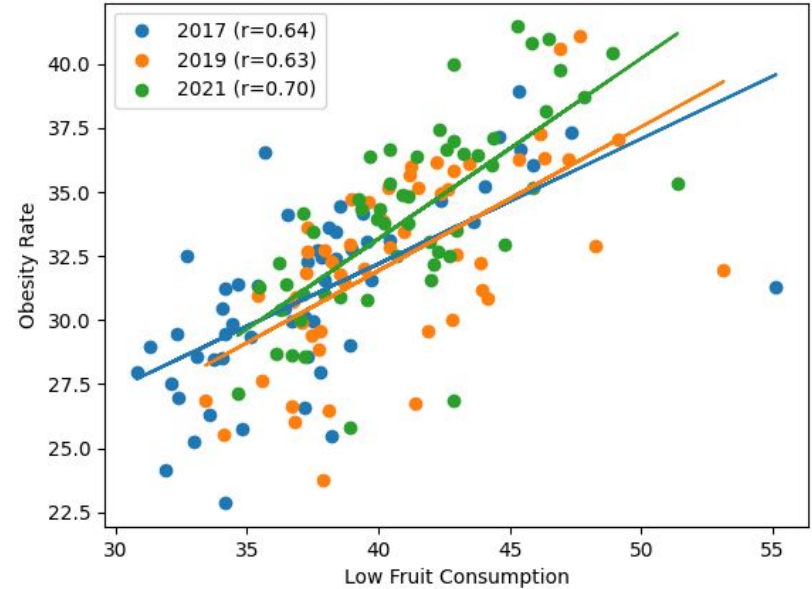
2011 MS Percent of adults aged 18 years and older who have obesity 39.9
2011 MS Percent of adults aged 18 years and older who have obesity 40.8
2011 MS Percent of adults aged 18 years and older who have obesity 37.5
2011 MS Percent of adults aged 18 years and older who have obesity 27.9
2011 MS Percent of adults aged 18 years and older who have obesity 31.2
2011 MS Percent of adults aged 18 years and older who have obesity 29.2
2011 MS Percent of adults aged 18 years and older who have obesity 37.3
2011 MS Percent of adults aged 18 years and older who have obesity 34.7
2011 MS Percent of adults aged 18 years and older who have obesity 26.8
2011 MS Percent of adults aged 18 years and older who have obesity 37.6
2011 MS Percent of adults aged 18 years and older who have obesity 35.4
2011 MS Percent of adults aged 18 years and older who have obesity 32.4
2011 MS Percent of adults aged 18 years and older who have obesity 42.9
2011 MS Percent of adults aged 18 years and older who have obesity 30.7
2011 MS Percent of adults aged 18 years and older who have obesity 36.9
2011 MS Percent of adults aged 18 years and older who have obesity 34.9
2011 MO Percent of adults aged 18 years and older who have obesity 34.2
```

Plots for Question 2

Obesity vs Physical Inactivity (by Year)



Obesity vs Dietary Habits (by Year)



Conclusions for Question 2

Key Findings

- The data showed a general relationship between physical activity, diet, and obesity rates, though not entirely conclusive.
- Individuals with lower levels of physical activity generally had higher obesity prevalence.
- Lower consumption of fruits and vegetables was also associated with increased obesity rates.
- This does present the issue of correlation vs. causation, so further analysis and refinement could be undertaken to better understand the relationships at play.



Selected Question 3

Selected Question

Is there a relationship between obesity rates across different demographic groups

Proposed Analysis

- Demographic averages
- Demographic comparisons using grouped bar charts
- Correlation analysis among variables

Expected Result

Demographic groups with higher incomes and higher education will have lower obesity rates

Metrics

- Income
- Education
- Race

Python Packages

- pandas
- matplotlib

Plots

- Bar graph

Question 3 – Command Line Output

```
(venv) PS C:\Users\Joshr\OneDrive\Desktop\ENGR\ENGR315\ENGR315-sp2026-student> c:\U
NGR\ENGR315\ENGR315-sp2026-student\venv\Scripts\python.exe "c:/Users/Joshr/OneDrive
-sp2026-student/Final Project/Q3withincome"
```

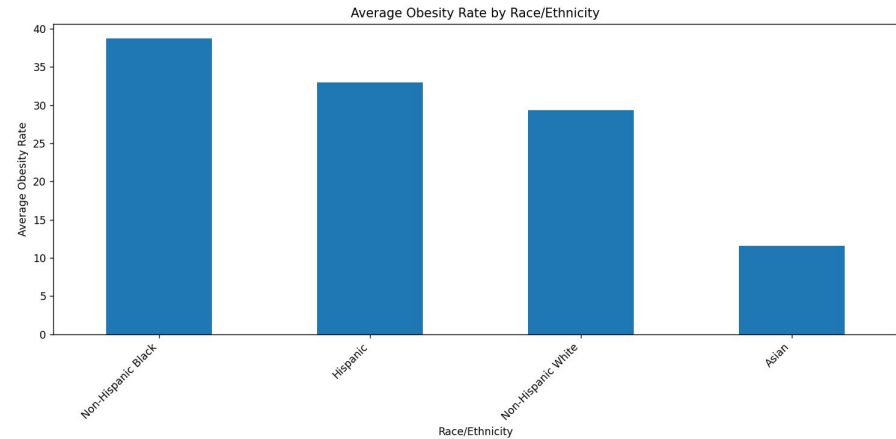
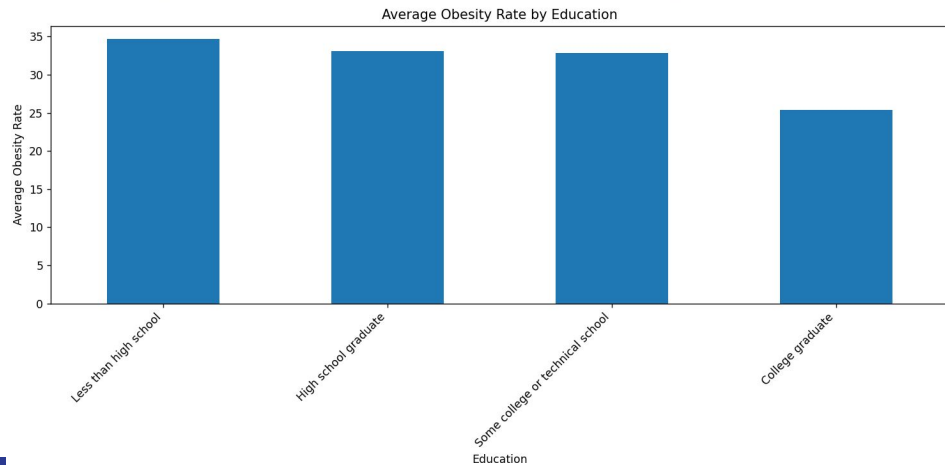
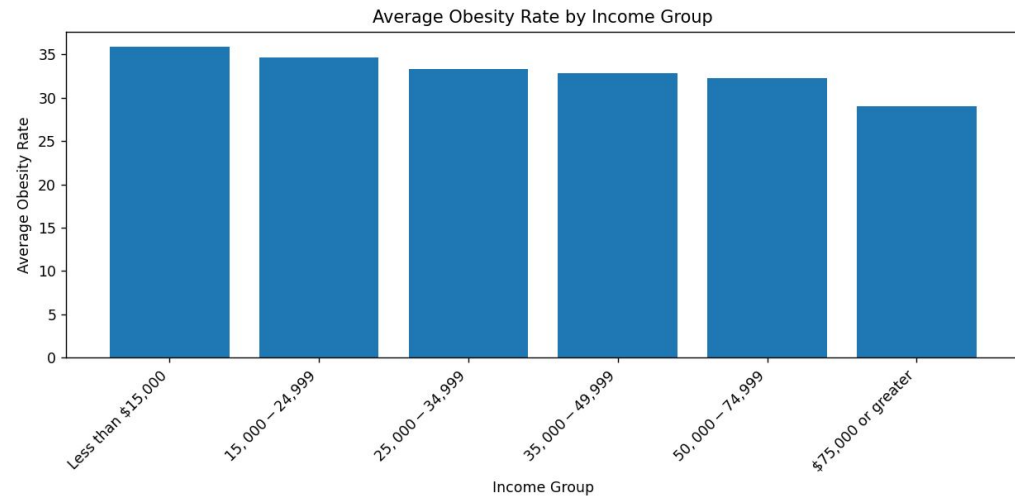
```
QUESTION: Percent of adults aged 18 years and older who have obesity
Demographic: $15,000 - $24,999 | Average Obesity Rate: 34.67
Demographic: $25,000 - $34,999 | Average Obesity Rate: 33.28
Demographic: $35,000 - $49,999 | Average Obesity Rate: 32.87
Demographic: $50,000 - $74,999 | Average Obesity Rate: 32.28
Demographic: $75,000 or greater | Average Obesity Rate: 29.02
Demographic: 18 - 24 | Average Obesity Rate: 18.54
Demographic: 2 or more races | Average Obesity Rate: 32.6
Demographic: 25 - 34 | Average Obesity Rate: 29.92
Demographic: 35 - 44 | Average Obesity Rate: 34.64
Demographic: 45 - 54 | Average Obesity Rate: 36.92
Demographic: 55 - 64 | Average Obesity Rate: 35.67
Demographic: 65 or older | Average Obesity Rate: 28.37
Demographic: American Indian/Alaska Native | Average Obesity Rate: 37.52
Demographic: College graduate | Average Obesity Rate: 25.38
Demographic: Data not reported | Average Obesity Rate: 26.82
Demographic: Female | Average Obesity Rate: 31.13
Demographic: High school graduate | Average Obesity Rate: 33.13
Demographic: Hispanic | Average Obesity Rate: 32.92
Demographic: Less than $15,000 | Average Obesity Rate: 35.84
Demographic: Less than high school | Average Obesity Rate: 34.67
Demographic: Male | Average Obesity Rate: 30.8
Demographic: Non-Hispanic Black | Average Obesity Rate: 38.71
Demographic: Non-Hispanic White | Average Obesity Rate: 29.31
Demographic: Some college or technical school | Average Obesity Rate: 32.85
Demographic: Total | Average Obesity Rate: 30.98
Demographic: Other | Average Obesity Rate: 28.1
Demographic: Asian | Average Obesity Rate: 11.6
Demographic: Hawaiian/Pacific Islander | Average Obesity Rate: 40.79
```

Average Obesity Rates by Income Group

```
Income Group: $15,000 - $24,999 | Average Obesity Rate: 34.67
Income Group: $25,000 - $34,999 | Average Obesity Rate: 33.28
Income Group: $35,000 - $49,999 | Average Obesity Rate: 32.87
Income Group: $50,000 - $74,999 | Average Obesity Rate: 32.28
Income Group: $75,000 or greater | Average Obesity Rate: 29.02
Income Group: Less than $15,000 | Average Obesity Rate: 35.84
```

```
(venv) PS C:\Users\Joshr\OneDrive\Desktop\ENGR\ENGR315\ENGR315-sp2026-student> █
```

Plots for Question 3



Conclusions for Question 3

1. Income vs. Obesity Rate

There is a clear negative trend:

- Lower income groups (e.g., “Less than \$15,000”) have higher average obesity rates (~35–36%)
- Higher income groups (e.g., “\$75,000 or greater”) have lower rates (~29–30%)

This suggests a moderate inverse correlation: as income increases, obesity rates tend to decrease.

2. Education vs. Obesity Rate

There is also a clear educational gradient:

- Lower education levels (e.g., “Less than high school”) show higher obesity rates (~35%+)
- Higher education levels (e.g., “College graduate”) show lower obesity rates (~28–31%)

This indicates a negative relationship between education level and obesity, where higher education is associated with lower obesity rates.

3. Race/Ethnicity vs. Obesity Rate

Differences are present but less uniform than income or education:

- Some groups (e.g., non-Hispanic Black and Hispanic populations) show higher average obesity rates (~35%+)
- Other groups (e.g., non-Hispanic White and Asian populations) show lower rates (~25–32%)

This suggests variation across racial/ethnic groups, but no single linear trend, indicating that obesity is influenced by multiple social and environmental factors beyond race alone.



Reflection on the Project

Overall, this project provided valuable experience in working with large datasets and applying Python-based data manipulation techniques. The size of the dataset, which would be impractical to analyze manually, was effectively managed and interpreted using computational tools. This demonstrated how programming can be leveraged as an engineering tool to improve data analysis capabilities, even in disciplines where software development is not the primary focus.





Reflections on Person Growth

Josh

At the start of the course, I had very limited experience with python programming and signal processing. Over the semester, I became much more confident with using python libraries like NumPy, SciPy, and Matplotlib to analyze data and build engineering applications. One of the biggest things I learned was how digital signal processing works in real applications. Through the ECG project, I learned how filtering, differentiation, and moving averages can improve noisy signals and make important features easier to detect. I also significantly improved my debugging and problem-solving skills. Testing the algorithm on multiple datasets was challenging, but showed how performance can change under different conditions and taught me the importance of evaluating the strengths and limitations of a system. Overall, this class was a good learning experience and I'm very happy that I have a good coding baseline so that it will be easier to learn other languages in the future.



Jake

At the beginning of this course I knew a little bit of python coding. I knew simple commands and how to perform If, When conditions as well as For, While loops. My knowledge was largely limited to simple mathematical formulas though. Processing large volumes of data and extracting certain information was way beyond my skill set. That skill is what I was hoping to get out of this course when I enrolled. For the most part, that skill was learned. I still struggle sometimes knowing which packages are the best to use for certain applications. The amount of options is somewhat crippling for me. I've never been a big computer guy and I think that has held me back a little bit with simple things in this course. For instance, knowing how to navigate the files through the terminal was an initial hurdle for me. I'm happy I've taken this course and do feel that the processing tools I've used will be useful for many applications professionally and personally. However I do feel that you can only get really good at these tools by consistently using them and I don't think I will do that. It's worth noting that a strong foundation of mathematics and statistics are necessary to understand this course.



Daniel

Overall, this course greatly improved my confidence and familiarity with programming as an engineering tool. Despite entering the course with no prior Python experience, I developed a strong understanding of fundamental concepts through assignments, labs, exams, and the final project. The course effectively demonstrated how Python can be applied to solve engineering problems and analyze large datasets that would be impractical to process manually, leaving me better prepared to use programming in future engineering work.

